

Shrivas Sudharsan

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EDUCATION

University of California San Diego | La Jolla, CA

2025-2029

B.S. in Artificial Intelligence | GPA: 4.0

- **Courses:** Discrete Mathematics, Physics, Calculus, Neuroscience, Macroeconomics, Machine Learning, Python

PROFESSIONAL EXPERIENCE

Artificial Intelligence Intern | *Photon Insights*

June 2024 - Aug. 2024

- Executed user-level **diagnostics** through interview-driven data collection and sentiment analysis, producing technical recommendations including the integration of a **proactive AI** feature.
- Worked with **LLM** developers and **quant** experts to improve accuracy of reports.
- Evaluated **machine learning** algorithms across multi-year financial datasets, competitor metrics, and news-driven data to enhance analytical precision.

Embedded Systems Research | *Pioneer Academics*

May 2024 - Aug. 2024

- Worked with a professor from UC Irvine to create an **embedded system** that alerts local residents of fire danger
- Trained **neural networks** with 200+ epochs to detect fire and smoke with **90.2%** accuracy
- Modeled the smart camera using a **raspberry pi** to communicate between smoke/visual sensors and local residents
- Developed expertise in visual processing, decision making, algorithmic design, and **IoT** engineering

Machine Learning Paper | *Cambridge Center for International Research*

Oct. 2023 - Dec. 2023

- Compared **16+ machine learning** model's effectiveness on diabetes detection using Python with a professor from the University of Oxford
- Leveraged **classification matrices** to analyze 5+ key features from ECG signals to improve precision of random forest classifier to **87.5%**
- Engineered mean, standard deviation, and time-domain metric after applying **normalization**, down-sampling, and **noise-filtering** to strengthen model performance.

PROJECTS

TritonHub, Full-Stack Web Platform | *SanD Hacks*

Jan. 2026 - Jan. 2026

- Built a **full-stack** platform aggregating Canvas LMS and Gmail data into a unified dashboard to streamline academic workflows
- Developed a **React.js** frontend with a dynamic state with a **Flask-based** API handling authentication
- Implemented scalable data pipelines and **RESTful** endpoints to integrate external services

LLM Training, Fine-Tuning & Retrieval Systems Project | *RapidFireAI*

Dec. 2025 - Jan. 2026

- Fine-tuned GPT-2 using Supervised Fine-Tuning (**SFT**) with **LoRA** adapters for customer-support generation, achieving ROUGE-L = **0.056** and eval loss = **1.15**, prioritizing sequence-level output quality over token-level loss
- Built a Retrieval-Augmented Generation (**RAG**) pipeline on a **FiQA** dataset optimizing chunk size, overlap, retrieval, and reranking parameters with an **MRR = 65.5%**
- Designed **hyperparallel** experimentation workflows using RapidFireAI's **APIs**, creating real-time metric visualization and early stopping of suboptimal runs across **SFT** and **RAG** experiments

TECHNICAL SKILLS

Languages: Python, Java, C++

Libraries + Frameworks: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, Matplotlib, React.js, Flask